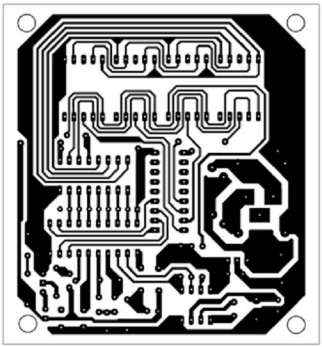


Fig. 3: An actual-size, single-side PCB for microcontroller-based clock using DS1307

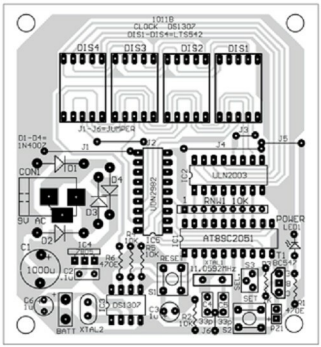


Fig. 4: Component layout for the PCB

Software

The source code is written in Assembly language and assembled using a cross-assembler (ASM51). It is well-commented and easy to understand.

The program uses internal Timer0 of the microcontroller for periodic data output to the LED displays. Timer0 is programmed as a 16-bit timer. The timer starts from F0FFH and overflows at FFFFH. On every overflow, its interrupt function is enabled. The interrupt service routine refreshes the display. For this, the minutes and hours data is read from the RTC and displayed on the seven-segment display.

For setting the time when switch S2 is pressed, the time is incremented and stored into the RTC. Set the alarm time by using switch S2 in combination with switch S3. The set time is stored in the RTC in two RAM locations. Every time the clock-time changes, it is compared with the stored alarm time value and if both are same, pin 7 of the microcontroller becomes high. Transistor T1 conducts and piezobuzzer PZ1 sounds for a duration of one minute.

EFY note. The source code of this project is available on www.efymag.com website.

The author retired as a professor from Alagappa College of Technology, Guindy in Chennai

CLOCK.ASM

```
$mod51
RB0     EQU     00H     ; Select Register
                          Bank 0
RB1     EQU     08H     ; Select Register
                          Bank 1
RB2     EQU     10H     ; Select Register
                          Bank 1

;%%%%%%%%%% %%%%%%%%%%
;       PORT DECLERATION
;%%%%%%%%%% %%%%%%%%%%
SDA     EQU     P3.1    ;SDA=PIN5
SCL     EQU     P3.0    ;SCL=PIN6
timeset EQU     P3.2    ; Port3,Bit2 is
used as timeset. A switch is wired to
press low.
alarset EQU     P3.3    ;Pin no.7 used
                          for alarm
DS1307W EQU     0D0H    ; SLAVE ADDRESS
                          1101 000 + 0 TO
                          WRITE
DS1307R EQU     0D1H    ; SLAVE ADDRESS
                          1101 000 + 1 TO
                          READ
segment_A EQU   P1.5
segment_B EQU   P1.4
segment_C EQU   P1.1
segment_D EQU   P1.2
segment_E EQU   P1.3
segment_F EQU   P1.6
segment_G EQU   P1.7
DIS4    EQU     P1.0
DIS3    EQU     P3.5
DIS2    EQU     P3.4
DIS1    EQU     P3.7
DSEG            ; This is internal data
                  memory
ORG     20H     ; Bit adressable memory
FLAGS1: DS      1
CONTROL BIT     FLAGS1.0 ; toggles with
                  every new keystroke
NEW     BIT     FLAGS1.1 ; Bit set when
                  a new command has been
                  received
FLAGS:  DS      1
LASTREAD BIT    FLAGS.0
SQW     BIT     FLAGS.4
ACK     BIT     FLAGS.5
BUS_FLT BIT     FLAGS.6
_2W_BUSY BIT    FLAGS.7
;byte values
BITCNT: DS      1
BYTECNT:DS      1
BYTE:   DS      10
BYTES:  DS      10
SECS:   DS      1       ;'SECONDS STORAGE
                          RAM
MINS:   DS      1       ;'   MINUTES    '
HRS:    DS      1       ;'   HOURS      '
DAY:    DS      1
DATE1:  DS      1
MONTH:  DS      1
YEAR:   DS      1
TOTAL:  DS      1       ;TOTAL COUNT
alahr:  DS      1
alamin: DS      1
TEMP:   DS      1
TEMPS:  DS      1
keypr:  DS      1
```